

Leveraging Workarounds for a Problem-Focused Improvement of Business Processes

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Appendix (available online via <http://link.springer.com>)

Appendices

Appendix 1

Table A: Cross-section of interviewees.

Org.	Interviewee	Role in the organization	Role in the process	Years in the organization	Relevant process
1	1	Customer manager	Initiator / Operation	21	Sales Forecast
	2	Customer manager	Initiator / Operation	4	
	3	Department manager	Approver	9	
	4	Orders operation	Operation	11	
	5	Orders operation	Operation	6	
	6	Presale engineer	Operation	8	
	7	Director of Procedures and Methods (P&M)	Supervisor	8	
	8	Director of Procurement and Logistics	Approver	3	
2	1	Chief Technology Officer (CTO)	Supervisor	3	Customer Complaint Management
	2	Lab and quality manager	Operation / Supervisor	16	
	3	Department manager	Initiator / Approver	23	
	4	Customer manager and nutritionist	Initiator	3	
	5	IS manager	Consultant	2	
3	1	CRM manager	Supervisor	10	Student Registration
	2	VP of Academic Administration	Approver	16	
	3	Head of the Academic Advisors Section	Operation	17	
	4	Registration coordinator	Initiator	2	
	5	Head of Information Center	Initiator / Approver	5	
	6	Head of Collection and Tuition Section	Operation	20	
	7	Academic advisors	Operation	8	
	8	Academic advisors	Operation	7	
	9	External consultant	Consultant	10	

Table B: Collected data.

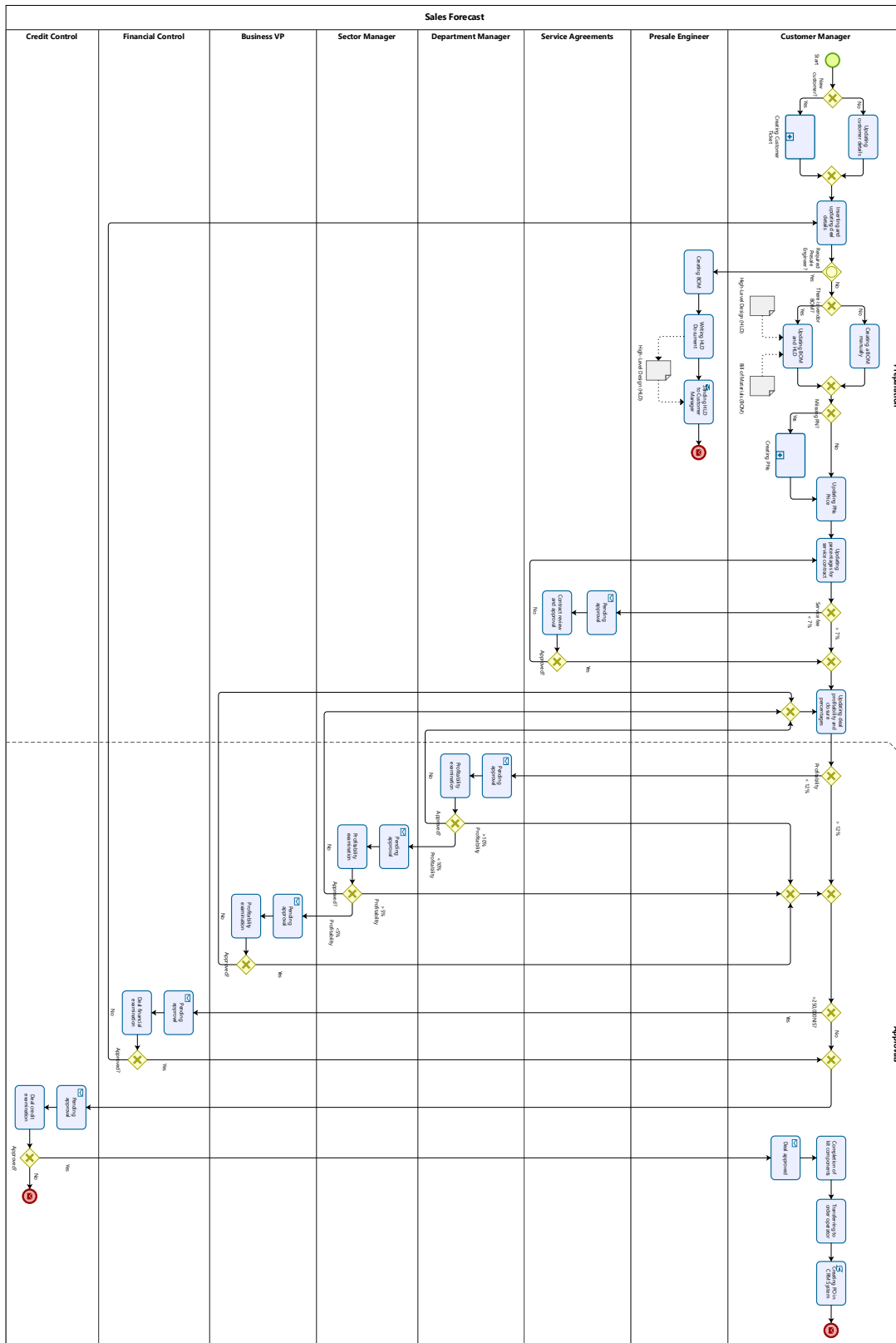
Process	Document type				Event log
	Formal procedures (ISO/SOX)	Implementation & characterization	As-is process models	IS Screenshots / reports	
Sales Forecast	2	1	2	13	46,098
Customer Complaint Management	1	1	1	7	23,109
Student Registration	2	1	1	11	30,017
Purchase Order	1	1	2	9	38,333

Table C: Guiding questions for the semi-structured interviews.

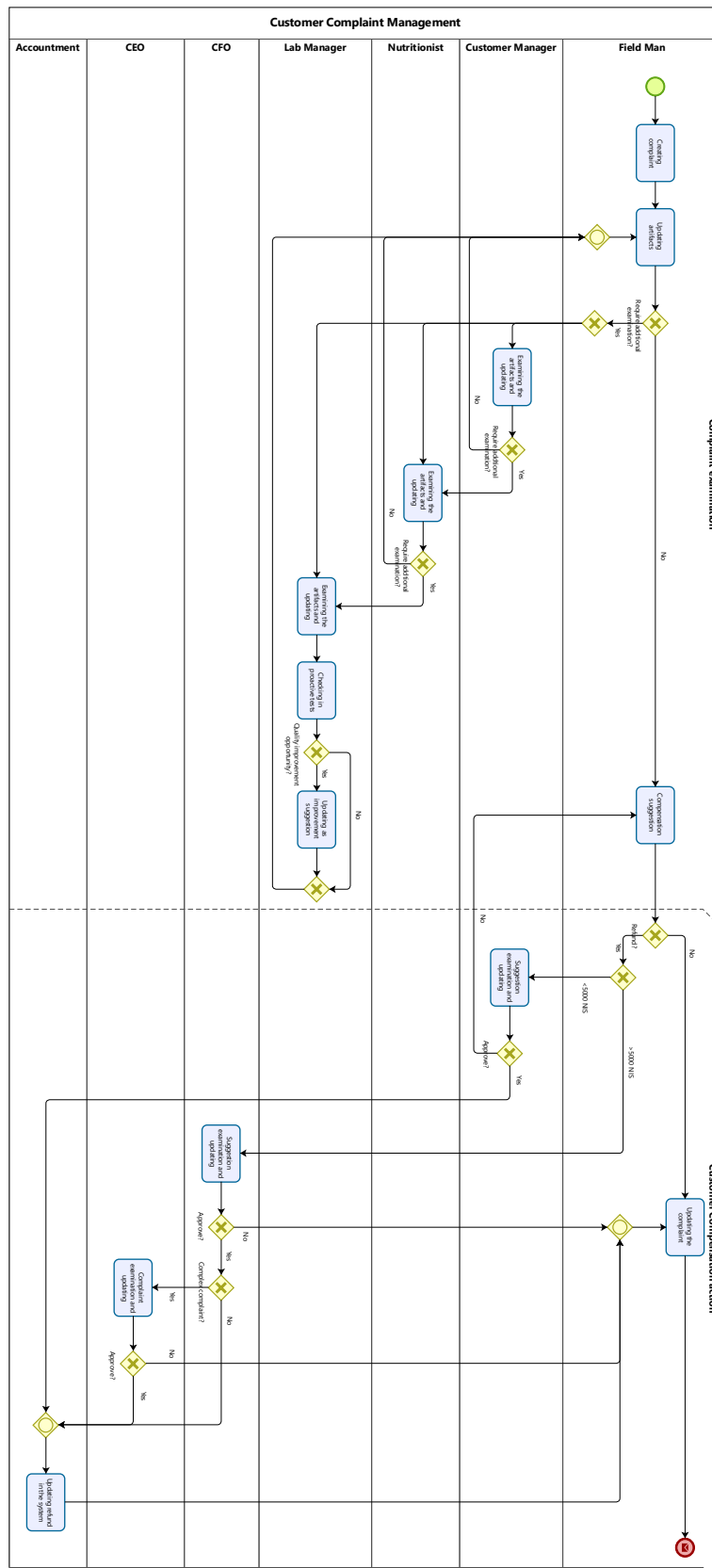
Interview questions	Goal	Used for
Organization and process		
Describe your role and area of responsibility in the organization.	Background and introduction.	Background knowledge
In your opinion, does the organization follow clear work procedures and standards like ISO, SOX, JCI? Are they implemented as expected?	To understand commitments to legal and external procedures, policies, and regulations.	Background knowledge
In your opinion, how are organizational and procedural changes managed and accepted?	To uncover change management issues, fixed paradigms, and resistance to changes.	Background knowledge
What are the organization's core processes? For each process, state whether it is: departmental, cross-organizational, or global?	To assess the studied process and examine its scope and boundaries.	Background knowledge
Describe the process from your viewpoint. What would you suggest in order to improve the process? Do you think the IS contributes to the effectiveness and efficiency of the process? Please elaborate.	To learn the process. To elicit desired goals of improvement ideas. To uncover misalignments between the actual process and the perceived goals.	- BPMN model - i* model - Conflict identification
What is your role in the process? Do you think your areas of responsibility in the process fit your area of expertise? How do they do so?	To elicit actor boundaries, tasks, goals and dependencies. To elicit workarounds, possibly those related to expertise.	- BPMN model - i* model - Workarounds list
Do you think that other process participants exceed their responsibility areas? How do they do so?	To uncover workarounds performed by others.	Workaround list
Is manual work included in the process? Where?	To uncover difficulties in the process and workarounds.	Workaround list
Participant's perceptions of goals and possible misalignments		
From your point of view, what are the organizational goals the process is intended to achieve? Were the process goals presented or explained to you?	To uncover perceived organizational goals. Possibly, to elicit related workarounds.	- i* model - Workaround list
What are the main goals you are expected to achieve as part of your role? In what ways are others involved and contribute to achieving these goals?	To uncover perceived local-unit goals. To learn the dependency relations among the actors.	- i* model - Conflict identification
What are the tasks you need to perform although they do not contribute directly to the goals you are expected to accomplish? Why do you need to perform them?	To elicit perceived misalignments between goals and tasks. To uncover motivation for workarounds.	- i* model - Conflict identification

Are there known and defined KPIs for the process? Are they measured on a personal or organizational level?	To elicit quality goals.	i* model
Are there certain situations that justify performing actions other than what is required in the process?	To examine and learn about goal misalignment situations in detail. To elicit workarounds.	- Conflict identification - Workaround list
In your opinion, does the official process include all the tasks required to accomplish your goals? If not, what do you think is missing?	To uncover perceived obstacles, gaps, and technological barriers in the process. To identify process improvement opportunities.	- Workaround list - Conflict identification
In general, are you satisfied with the process? Would you suggest other ways of achieving the relevant goals?	To examine improvement ideas, including those already performed informally (as workarounds).	Workaround list
Workarounds in the process		
In your opinion, what is a business process workaround? Do you think management would define it similarly? If not, how do you think they would define it?	To assess perceptions about management's view of workarounds.	Background knowledge
Are you familiar with workarounds that are performed in the process? If so, in which situations and how often? How are they identified?	To examine and learn about workarounds in detail, their frequencies, and the impact to the organization.	Workaround list
Why, in your opinion, are these workarounds performed? Do these workarounds help participants accomplish their goals? Do you think that these workarounds help accomplish organizational goals?	To learn the motivation for process workarounds. To relate the misalignment situations to specific workarounds.	- Workaround list - Conflict identification
What do you think are the consequences of these workarounds? What would happen if they could not be performed?	To elicit perceived impacts of workarounds.	Conflict identification
Do you think that the process workarounds are related to obstacles of any kind? Do you think that with a more flexible process they would still be performed?	To understand the motivation for performing workarounds To examine process improvements / opportunities.	Conflict identification
In your opinion, are workarounds performed regularly? Do they require skills or deep familiarity with the process?	To clarify the motivation and the effort involved in workarounds.	Workaround list

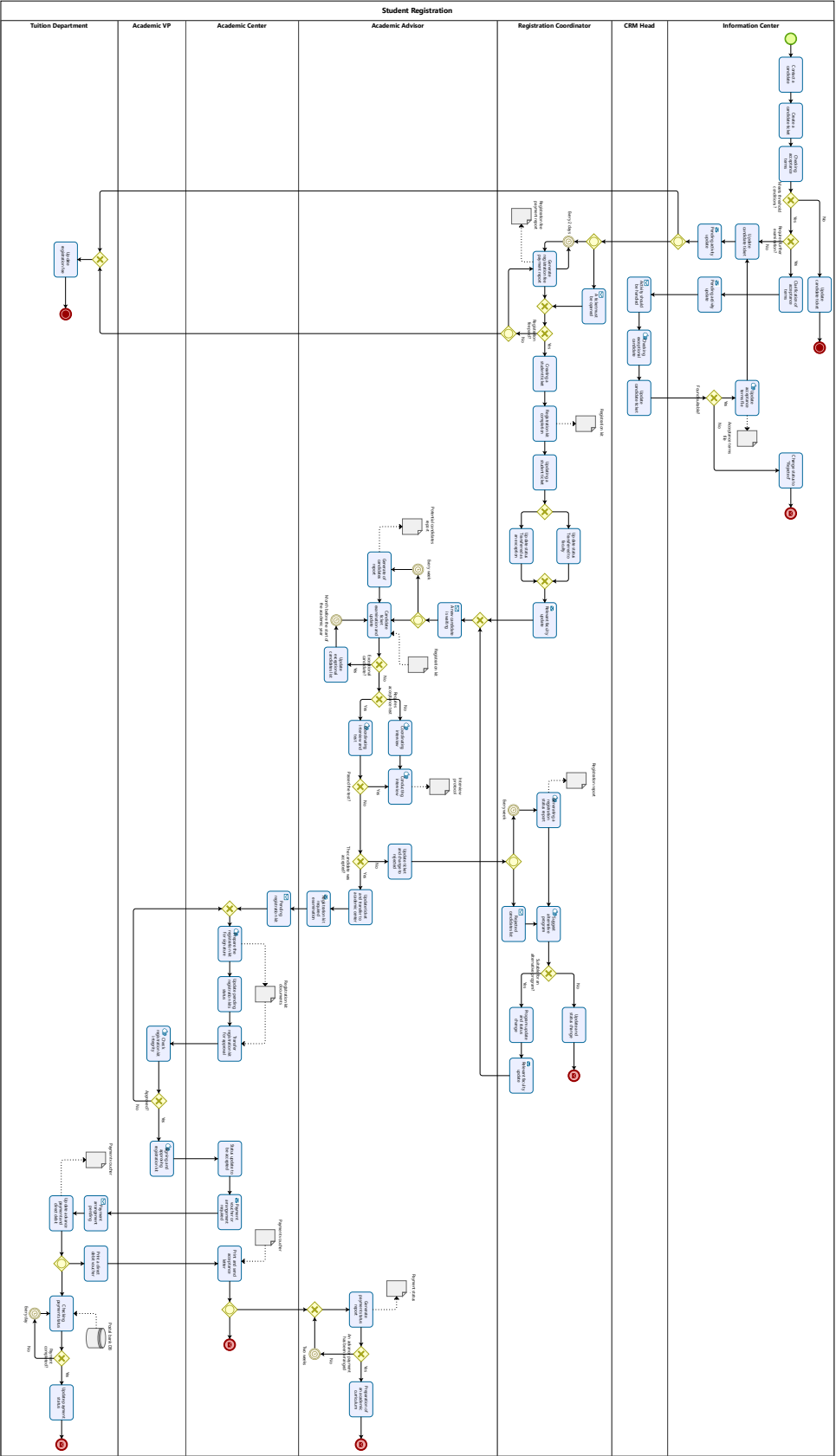
Process 1: Sales Forecast.



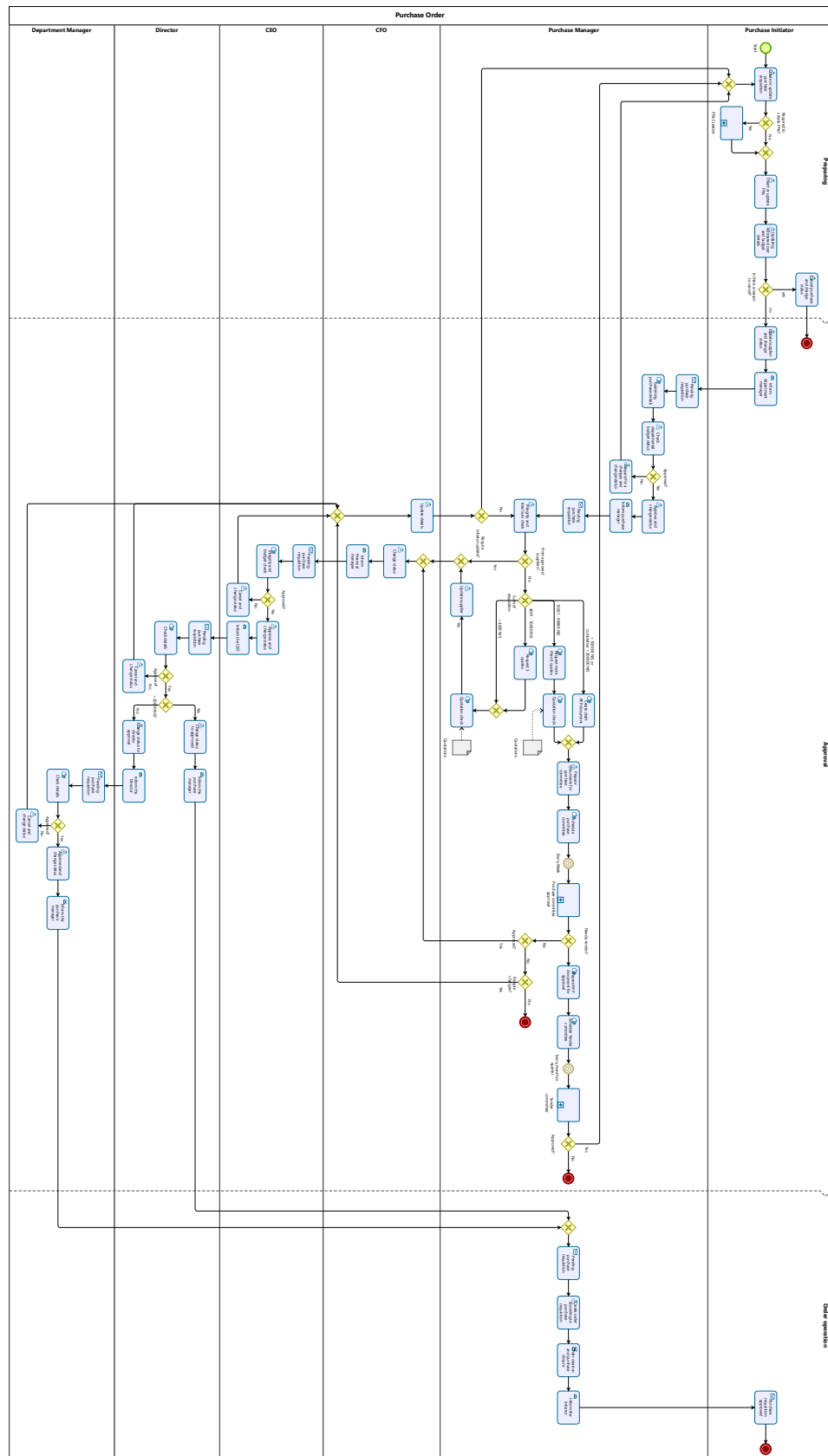
Process 2: Customer Complaint Management.



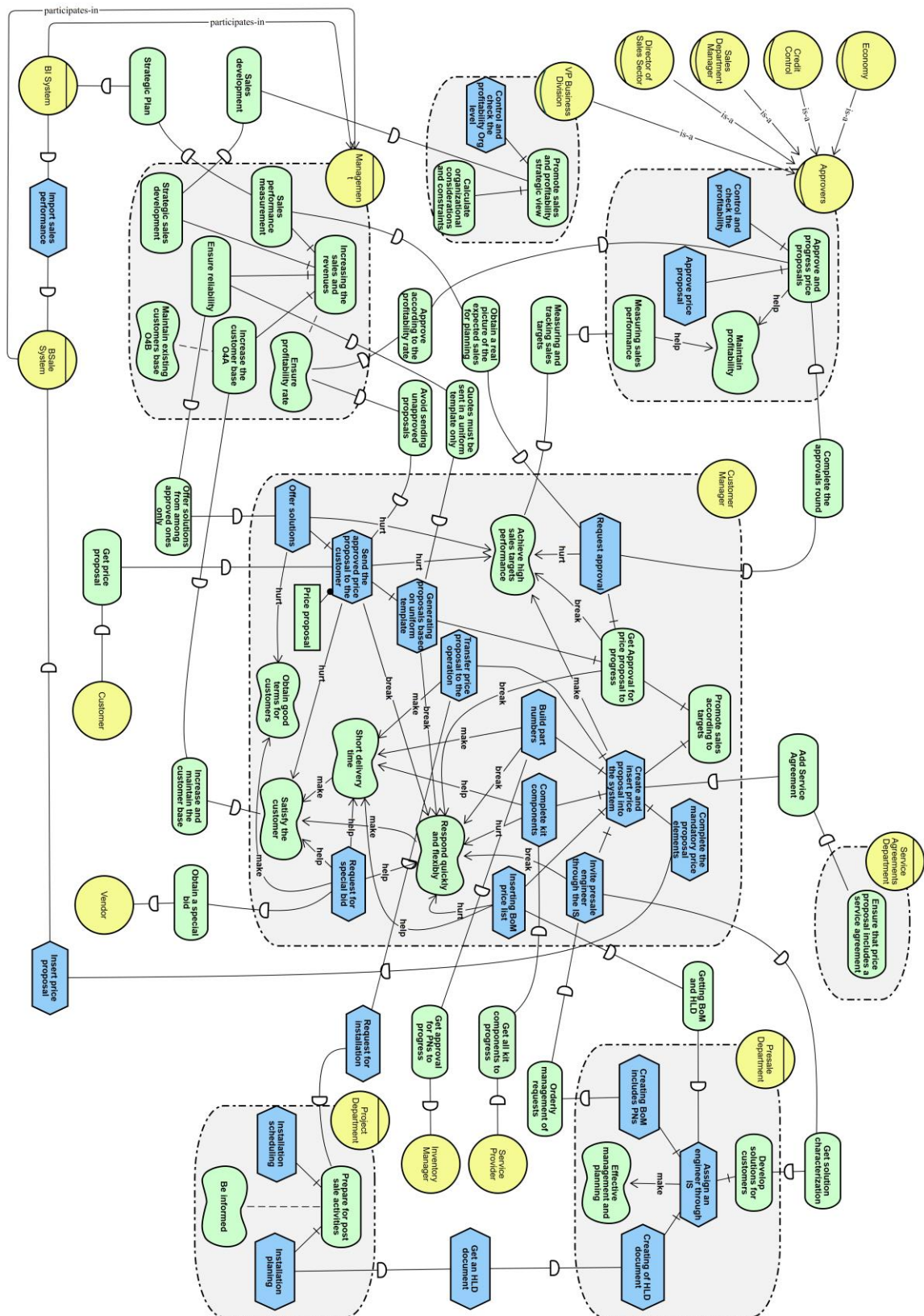
Process 3: Student Registration.



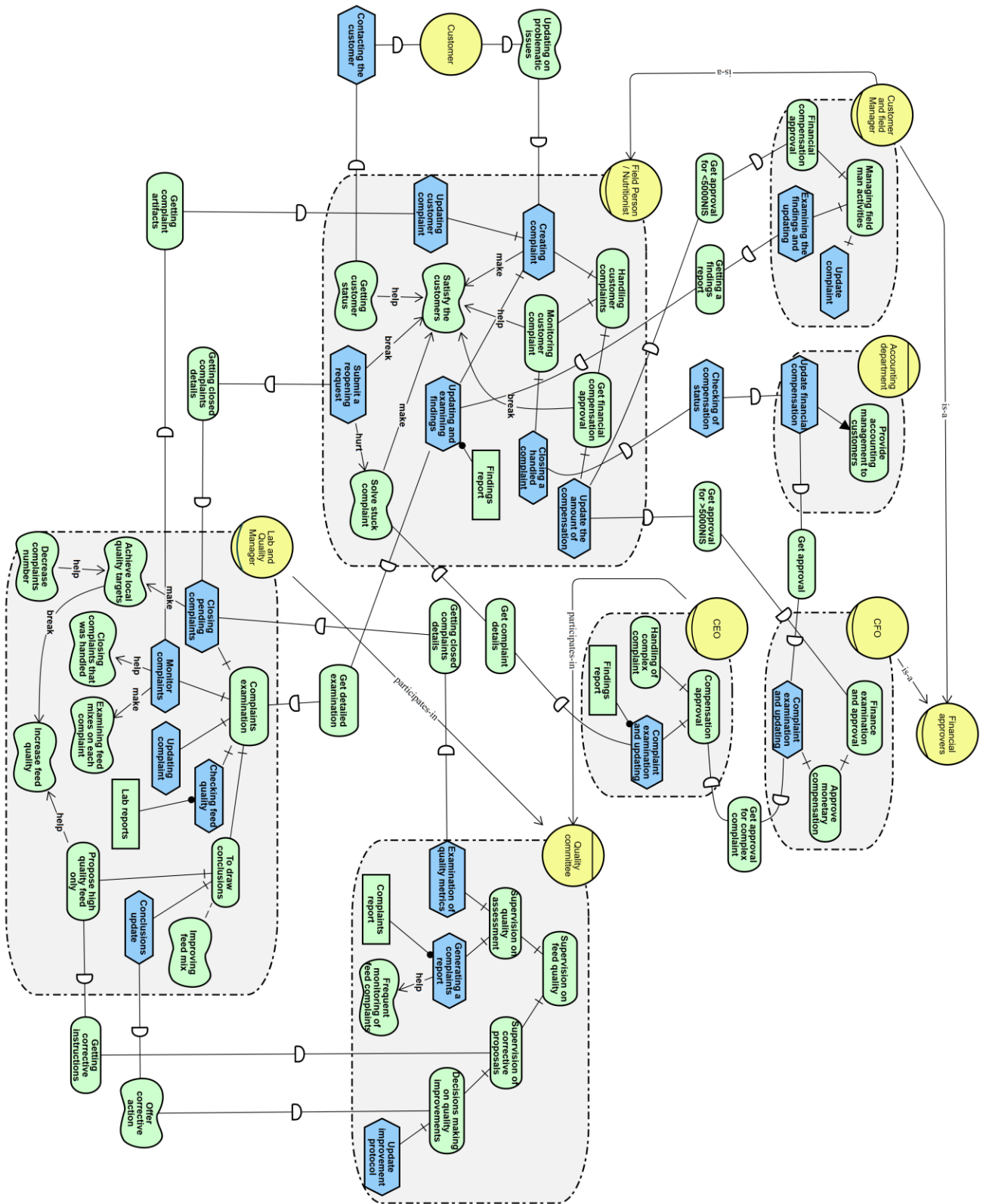
Process 4: Purchase Order.



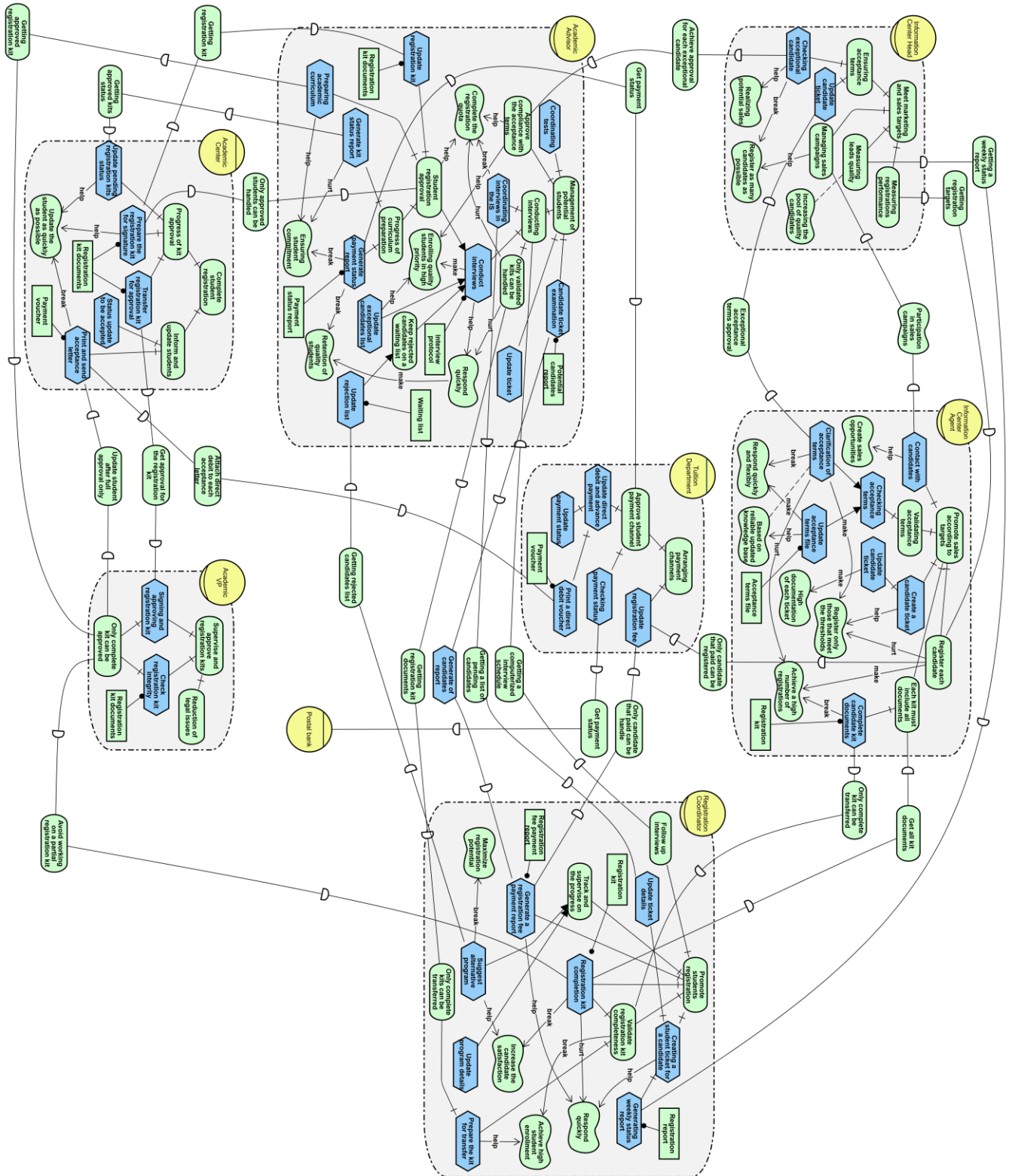
Process 1: Sales Forecast.



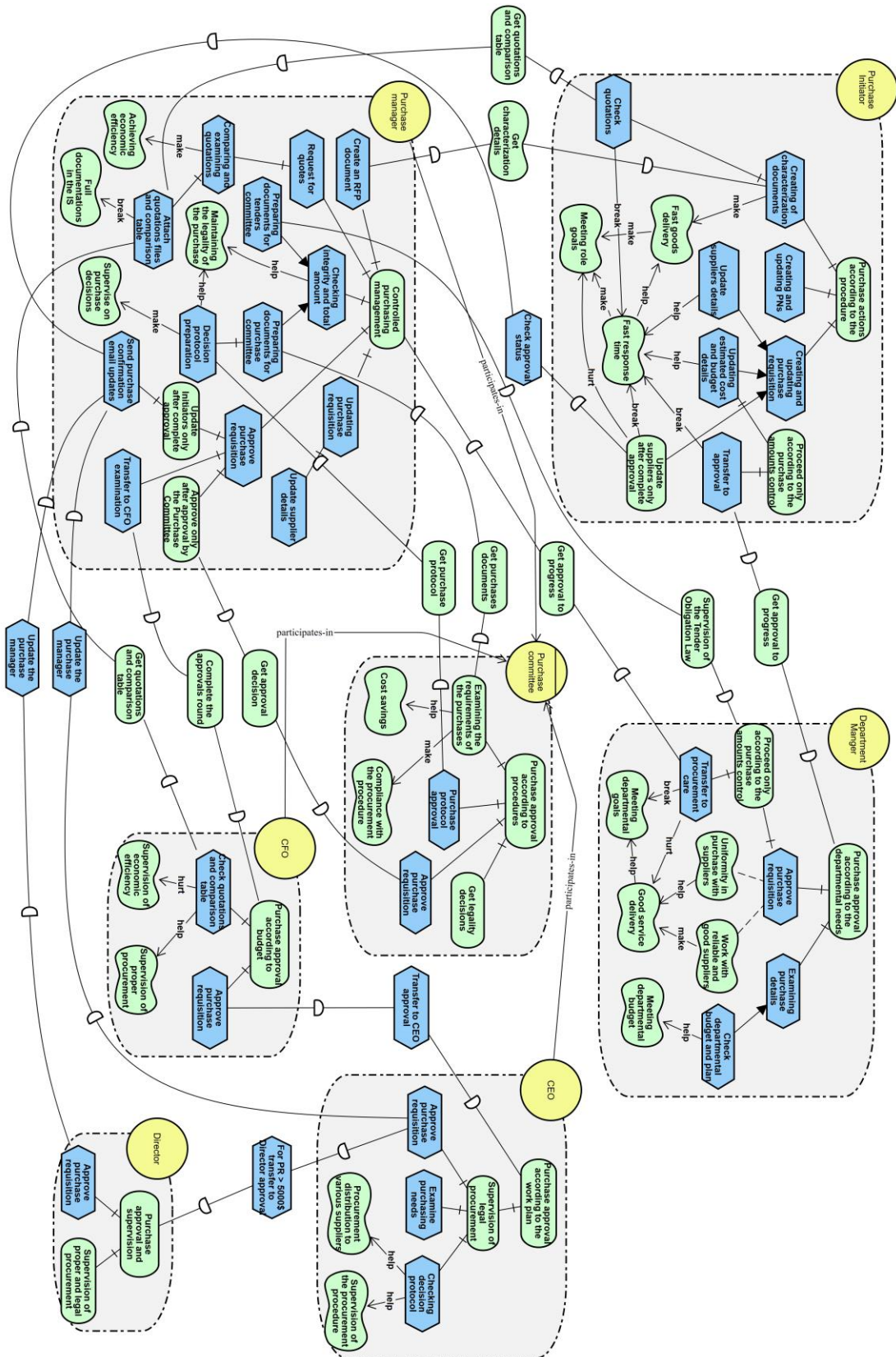
Process 2: Customer Complaint Management.



Process 3: Student Registration.



Process 4: Purchase Order.



Appendix 4: Misalignment situations.

Table A: Sales forecast process.

SN	Actor goals dependencies		Misalignment description
	Management	Customer manager	
1.1	Obtain a real picture of the expected sales for planning	Achieve high sales targets performance	The customer managers are evaluated and rewarded on closing deals only, while the management desire to get a real picture of the sales activities, regardless of closures.
1.2	Avoid sending unapproved proposals	Satisfy the customer	To mitigate organizational risks, the organizational management issues instructions on avoiding the sending of unapproved price proposals. However, obtaining approval for each price proposal and any subsequent modifications can introduce delays that hinder the customer manager's ability to respond quickly and satisfy the customer. This, in turn, may also affect the achievement of the required sales targets.
1.3		Respond quickly and flexibly	
1.4		Achieve high sales targets performance	
1.5	Offer solutions from among approved ones only	Obtain good terms for customers	Limiting customer managers to offering solutions included in a closed list only restricts their ability to provide flexible solutions with favorable terms to customers, which their competitors are able to do. Consequently, this constraint also hinders the attainment of their local sales targets.
1.6		Achieve high sales targets performance	
1.7	Quotes must be sent in a uniform template only	Respond quickly and flexibly	The organizational management's goal of utilizing the uniform template, as currently configured in the process, does not align with the specific needs and demands faced in the field. This misalignment imposes limitations on the adaptability and responsiveness of customer managers to meet their customers' requirements.
SN	Approvers	Customer manager	Misalignment description
1.8	Get approval for a price proposal to progress	Respond <u>quickly</u> and flexibly	As detailed in Situations 1.2, 1.3, and 1.4, the necessity for a quote approval trail is recognized as a risk mitigation measure. However, within the context of sales activities, closing a deal often necessitates multiple iterations with the customer. Requiring a new approval for each iteration hinders the customer managers' ability to respond promptly to the customer's needs and ultimately affects their capacity to achieve the desired sales targets.
1.9		Achieve high sales targets performance	
SN	Inventory manager	Customer manager	Misalignment description
1.10	Get approval for PNs to progress	Respond quickly and flexibly	Each quotation necessitates the early-stage approval of all PNs, even though these may evolve during negotiations with customers. This mandatory approval process introduces delays that hinder customer

			managers from responding as promptly as is typically expected of them.
SN	Service provider	Customer manager	Misalignment description
1.11	Get all kit components to progress	Respond quickly and flexibly	Each quote necessitates in the process early-stage approval of all kit components, including elements such as business-signed customer documents, even though they are not typically required at that stage. Compelling the completion of the entire set of kit documents at these early stages results in delays that impede customer managers from responding promptly, as is typically expected of them.
SN	Presale Department	Customer manager	Misalignment description
1.12	Orderly management of invitation requests	Respond quickly and flexibly	As an integral part of the process, customer managers rely on the support of presale engineers to define solutions for customers. However, because of the incompatibility of the invitation feature, those who have used this feature cannot secure the assistance of an engineer within the necessary timeframe. This issue adversely affects the ability of customer managers to respond promptly to customer needs.
1.13	Getting BoM and HLD	Respond quickly and flexibly	The customer managers are required to include in each quote detailed characterization documents in the early stages of the process. Because of Situation 1.12, and since this document does not require an early stage (and also may change), the customer managers cannot respond quickly as expected of them.

Table B: Customer complaint management process.

SN	Actor goals dependencies		Misalignment description
	Lab and Quality Manager	Field Person / Nutritionist	
2.1	Closing pending complaints	Satisfy the customers	Closing pending complaints without handling or updating them, even though the process permits it, can negatively impact the field persons' ability to satisfy their customers. This is because the complaint remains unresolved and may necessitate reopening.
2.2	Achieve local quality targets	Solve stuck complaint	The local quality targets, which are based on the number of open complaints, motivate the lab manager to close complaints. However, this action may have negative consequences for the field persons, as it may hinder their ability to address complaints that have been awaiting a response for an extended period.
SN	Lab and Quality Manager	Quality committee	Misalignment description
2.3	Achieve local quality targets	Increasing feed quality	As described in Situation 3.2 above, this behavior also creates a misalignment with the organizational goal of enhancing feed quality by thoroughly examining complaints, drawing conclusions, and subsequently implementing improvements.
SN	Financial approvers	Field Person / Nutritionist	Misalignment description
2.4	Get financial compensation approval	Satisfy the customers	Having to obtain financial approval for any amount of monetary compensation may impede the field persons' ability to satisfy their customers, as this approval process introduces delays in response times.

Table C: Student registration process.

SN	Actor goals dependencies		Misalignment description
	Information Center Head	Information Center Agent	
3.1	Ensuring acceptance terms	Respond quickly and flexibly	Ensuring acceptance terms for each candidate can be a time-consuming task that may depend on other actors. However, this may hinder the information center agent from responding quickly, as expected of them, and from meeting their registration number targets.
3.2		Achieve a high number of registrations	
3.3	Getting registration targets	Register only those that meet the thresholds	Registering only those candidates who meet the specified thresholds can be a complex goal when the agent aims to meet their quantitative registration targets.
SN	Information Center Head	Academic Advisor	Misalignment description
3.4	Register as many candidates as possible	Approve compliance with the acceptance terms	Approving compliance with the acceptance terms for each candidate may hinder the achievement of a high registration number.
SN	Information Center Agent	Registration Coordinator	Misalignment description
3.5	Achieve a high number of registrations	Validate registration kit completeness	Completing the registration kit documents using manual tools such as fax machines and scanners, as currently practiced, can impede the attainment of a high registration number.
3.6	Only complete kit can be transferred	Achieve high student enrollment	High student enrollment cannot be achieved when the registration coordinator must wait for complete kits only.
SN	Academic VP	Registration Coordinator	Misalignment description
3.7	Only complete kit can be approved	Respond quickly	As per the explanation in Situation 2.5, the registration coordinator cannot respond as anticipated and candidate satisfaction may be harmed until the kit is completely filled out.
3.8		Increase the candidate satisfaction	
SN	Registration Coordinator	Academic Advisor	Misalignment description
3.9	Only complete kits can be transferred	Respond quickly	Since the validation of the kit documents is very complex and time-consuming, waiting for full kit validation may delay the response of the academic advisors and may hinder them from achieving their local goals regarding the faculty registration quotas.
3.10		Complete the registration quota	
3.11	Maximize registration potential	Keep rejected candidates on a waiting list	Since the academic advisor opts to maintain rejected candidates on an external waiting list that may or may not be utilized in specific situations, the registration coordinator is unable to fully optimize the registration potential for these candidates.

3.12	Follow up interviews	Complete the registration quota	The registration coordinator can monitor only the status of interviews that have been coordinated within the process. However, because of limitations in the process capabilities, interviews cannot be entirely managed within the process. Consequently, academic advisors encounter challenges in meeting the registration quotas when they utilize the process for interview coordination.
SN	Tuition Department	Academic Advisor	Misalignment description
3.13	Approve student payment channel	Ensuring student commitment	Because the tuition department is responsible for processing payments for all students, including both new and existing ones, without any prioritization, new students experience delays and are unable to proceed in the process and cannot progress in the preparation of the curriculum. This, in turn, poses a challenge for academic advisors in retaining students who have already been accepted and in ensuring their commitment to the academic institution, which can contribute to achieving their registration targets.
3.14		Retention of quality students	
SN	Academic Center	Academic Advisor	Misalignment description
3.15	Progress of kit approval	Ensuring student commitment	To ensure a student's commitment, the student must complete the payment (as described in Situations 2.13 and 2.14) and prepare the academic curriculum. However, these steps cannot be taken until the kit is fully approved by the academic center.
SN	Academic VP	Academic Center	Misalignment description
3.16	Only integrity and complete kit can be approved	Update the student as quickly as possible	To avoid a scenario in which a student enrolls elsewhere, it is crucial to promptly inform the student of the completion of their registration. However, this cannot be accomplished until the kit is fully approved.

Appendix 5

Table A: Cross-section of BPI expert interviewees.

BPI Expert	Areas of experience	Part of the activities	Years of experience
1	Implements and develops ERP and BI systems. Owner of BI company	Implemented and developed dozens of business processes in an ERP environment	17
2	Owner and CEO of ERP company. VP of Technology at a security company.	A variety of business process implementations in a variety of environments, some in ERP and some in various ISs, including security and service systems	15
3	Analyzes and develops information systems including project management. Academic researcher in the BPM field.	A variety of developments and implementations of business processes in various environments. Publication of academic papers in BPM.	6
4	A product manager at a startup company that manages IS for production floors in industrial plants, with 4 years' experience. Previously, 8 years' experience as a production engineer in the army. Academic researcher in the BPM field.	Developed, implemented, and improved various business processes in the production industrial field	12
5	CTO at an industrial company with 10 years of experience. Owner of an ERP and CRM implementation company and consulting services . Academic lecturer with 20 years of experience in the information systems field; academic researcher	Extensive experience in the field of technology and information systems in management, consulting, and implementation positions, with dozens of business process implementations and improvements	22
6	Technology and information operator at Pharma company. Academic researcher in IS field.	Management, analysis, improvement, and implementation of logistics and planning processes	9
7	Leads efficiency and optimization projects in local and global organizations through measurement and control. Strategic consultant to organizations in a variety of fields and sectors, specializing in process and system analysis to improve and optimize processes through measurement, control, and analytics.	About 5 years' experience, working independently in consulting and improving business processes against the organizational strategy. Before, worked in several different companies including BI and strategic consulting in the field of business processes at the national and international levels. Includes companies in Asia, Europe, and the United States in a variety of fields such as CRM, BI, Bigdata, etc.	15

Table B: Structured interview with BPI experts.

QN	Question	Options	
1	Do you think this situation is related to a conflict or some other reason? What do you think this situation stems from? Please elaborate.	Free text	
2	What can be suggested to improve the process in this situation? Especially in order to reduce the conflict (if any)?	Free text	
3	Which conflict category can best suit and describe this situation? (Choose all that apply)	1	Conflicts arising from implemental / practical gaps in organizational procedures and / or regulation with respect to the field.
		2	Conflicts arising from implemental / technological gaps in the process or IS to support the required level of performance
		3	Conflicts arising from lack of adjustments or inconsistency in performance assessment or indicators of the process actors
		4	Others - Please elaborate (free text)
4	Which improvement category (one or more) can be proposed in order to reduce the conflict? (Choose all that apply)	1	Introduce or adjustment of KPI or business metrics
		2	Changes in process control gateways and their transition conditions
		3	Changes in process paths
		4	Increasing the flexibility level of the process and its activities
		5	Changes in responsibility boundaries and specialist
		6	Split or merge process tasks
		7	Improvements and changes to IS screens, features, and capabilities
		8	Adjustments of permissions level in the process
		9	Adjustments or implementation of service levels agreement (SLA)
		10	Improvements and changes in the process monitors, indicators, and alerts

Table C: Practical interview questionnaire.

QN	Question	Goals
1	From your practical experiences, how would you analyze the process and its situations in order to improve it?	To allow the experts to be open about their practical experience as a basis for the next questions.
2	How, if at all, can the proposed framework help you in analyzing business processes? Please elaborate	To assess the usefulness and the effectiveness of the framework.
3	What would you suggest in order to improve the framework? Elaborate on the practical aspects.	To reveal needed improvements and adjustments of the framework to practice.